Data Visualization Guide

Types of Graphs and Their Usages

Subject: Data Science Lab | Covers Experiments 1, 2 & 3 | Dataset: Titanic

1. Introduction

Data visualization converts numbers into pictures, making patterns, trends, outliers, and relationships easier to understand. Choosing the right graph depends on the type of variable (categorical vs numerical) and the question you want to answer.

Variable Type	Best Graph Options
Categorical	Bar chart, Count plot, Stacked bar, Pie chart
Numerical (one variable)	Histogram, Box plot
Numerical (two variables)	Scatter plot, Heatmap (correlation)
Two categorical	Stacked/grouped bar, Contingency heatmap
Many numerical	Correlation heatmap, PCA scatter

2. Types of Graphs (Detailed)

1. Bar Chart (Count Plot)

A chart with rectangular bars whose heights represent the frequency or count of each category.

When to use:

- Compare counts across categories
- Show distribution of a categorical variable
- Compare groups (e.g. male vs female passengers)

Aspect	Detail
Data type	Categorical variable (sex, class, survived, embarked)
What it shows	How many observations fall in each category
Python (Seaborn/Matplotlib)	sns.countplot() / df.value_counts().plot(kind='bar')
Used in experiment	Exp 1, Exp 2
Titanic example	Count of passengers by class (First, Second, Third) or by sex

2. Stacked Bar Chart

A bar chart where each bar is divided into segments showing sub-category proportions.

When to use:

- Compare composition of two categorical variables
- Show part-to-whole relationship within each group
- Compare survival rates across passenger classes

Aspect	Detail
Data type	Two categorical variables (e.g. class + survived)
What it shows	How one categorical variable is distributed within levels of another
Python (Seaborn/Matplotlib)	pd.crosstab(...).plot(kind='bar', stacked=True)
Used in experiment	Exp 1

Titanic example	Survival distribution by passenger class - survived vs not survived stacked

3. Histogram

A bar chart for numerical data showing frequency in continuous bins (intervals).

When to use:

- Understand distribution of a numerical variable
- Check skewness (symmetric, left, or right skewed)
- Identify central tendency and spread of Age or Fare

Aspect	**Detail**
Data type	Single numerical variable (age, fare)
What it shows	Shape of distribution - where most values cluster
Python (Seaborn/Matplotlib)	sns.histplot() / plt.hist() with bins and optional KDE curve
Used in experiment	Exp 2
Titanic example	Histogram of Age showing peak around 20-30 years; Fare right-skewed

4. Box Plot (Box-and-Whisker Plot)

Shows median, quartiles (Q1, Q3), and outliers using a box and whiskers.

When to use:

- Summarize numerical data in five numbers
- Detect outliers visually
- Compare distributions across groups

Aspect	**Detail**
Data type	Numerical variable (fare, age); optionally grouped by category
What it shows	Median (line in box), IQR (box width), min/max whiskers, outlier points
Python (Seaborn/Matplotlib)	sns.boxplot() / df.boxplot()
Used in experiment	Exp 2
Titanic example	Fare box plot - 116 outliers above upper fence; median << mean (right skew)

5. Scatter Plot

Plots individual data points using two numerical axes (x and y).

When to use:

- Examine relationship between two numerical variables
- Detect correlation, clusters, or trends
- Visualize PCA results (PC1 vs PC2)

Aspect	Detail
Data type	Two numerical variables (age vs fare, PC1 vs PC2)
What it shows	Pattern of co-variation; can color by a third variable (survival, species)
Python (Seaborn/Matplotlib)	sns.scatterplot(x=..., y=..., hue=...)
Used in experiment	Exp 1, Exp 2, Exp 3
Titanic example	Age vs Fare (weak correlation); PC1 vs PC2 colored by survival

6. Heatmap (Correlation Heatmap)

Color-coded matrix where cell color intensity represents value magnitude.

When to use:

- Show correlation between multiple numerical variables
- Identify strongly related feature pairs
- Visualize correlation matrix from df.corr()

Aspect	Detail
Data type	Numerical variables - correlation matrix or contingency table
What it shows	Strength and direction of relationships (red=negative, blue=positive)
Python (Seaborn/Matplotlib)	sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
Used in experiment	Exp 1, Exp 2, Exp 3
Titanic example	Correlation among age, fare, sibsp, parch, pclass; Age-Fare r=0.096

7. Line Chart

Connects data points with lines, usually over an ordered x-axis.

When to use:

- Show trend over time or ordered categories
- Plot cumulative explained variance in PCA
- Display time series data

Aspect	Detail
Data type	Ordered x (time, component number) and numerical y
What it shows	Trend, increase/decrease, rate of change
Python (Seaborn/Matplotlib)	plt.plot() / sns.lineplot()
Used in experiment	Exp 3
Titanic example	Cumulative explained variance vs number of PCA components

8. Pie Chart

Circle divided into slices proportional to category frequencies.

When to use:

- Show simple part-to-whole for few categories (2-5)
- Present percentage breakdown to non-technical audience

Aspect	Detail
Data type	Single categorical variable
What it shows	Proportion of each category as a slice
Python (Seaborn/Matplotlib)	plt.pie() - use sparingly; bar charts often clearer
Used in experiment	General (not primary in our labs)
Titanic example	Survived vs not survived as percentages - bar chart preferred in EDA

3. Quick Comparison Table

Graph	Variables	Main Purpose	Experiment
Bar Chart	1 categorical	Compare category counts	Exp 1, 2
Stacked Bar	2 categorical	Composition within groups	Exp 1
Histogram	1 numerical	Show distribution shape	Exp 2
Box Plot	1 numerical	Median, IQR, outliers	Exp 2
Scatter Plot	2 numerical	Relationship between vars	Exp 1, 2, 3
Heatmap	Many numerical	Correlation matrix	Exp 1, 2, 3
Line Chart	Ordered + numerical	Trend over order/time	Exp 3
Pie Chart	1 categorical	Part-to-whole proportions	General

4. How to Choose the Right Graph

- Want counts of categories? -> Bar chart / Count plot
- Want distribution of one number? -> Histogram or Box plot
- Want relationship between two numbers? -> Scatter plot
- Want correlation among many numbers? -> Heatmap
- Want composition of two categories? -> Stacked bar chart
- Want to detect outliers? -> Box plot
- Want PCA / dimensionality reduction view? -> Scatter plot (PC1 vs PC2)
- Want trend across ordered values? -> Line chart

5. Graph Usage by Experiment

Experiment 1 - Frequency, Probability & Correlation

- Bar chart: absolute frequency of passenger class
- Stacked bar chart: survival by class
- Heatmap: correlation between Age and Fare
- Scatter plot: Age vs Fare relationship

Experiment 2 - EDA & Descriptive Statistics

- Bar chart: sex, class, survived counts
- Histogram: Age and Fare distributions
- Box plot: Fare outliers (116 above upper fence)
- Scatter plot: Age vs Fare colored by survival
- Heatmap: correlation of all numerical variables

Experiment 3 - PCA

- Scatter plot: PC1 vs PC2 (2D visualization after reduction)
- Bar chart: explained variance per component
- Line chart: cumulative explained variance

6. Box Plot - Key Elements

Understanding box plot components (important for viva):

Element	Meaning
Box	Middle 50% of data (Q1 to Q3)

Line inside box	Median (Q2)
Whiskers	Extend to 1.5 x IQR from Q1 and Q3
Dots/points	Outliers beyond whiskers
IQR	Q3 - Q1 (interquartile range)

7. Viva Q&A

Q: When do you use a bar chart?

A: To compare counts or frequencies of categorical variables.

Q: Bar chart vs histogram?

A: Bar = categories (discrete). Histogram = numerical bins (continuous).

Q: When do you use a box plot?

A: To show median, quartiles, and detect outliers in numerical data.

Q: When do you use a scatter plot?

A: To examine relationship between two numerical variables.

Q: When do you use a heatmap?

A: To visualize correlation or values in a matrix form.

Q: What is a stacked bar chart used for?

A: To show how one categorical variable splits within another.

Q: Which graph detects outliers?

A: Box plot (points beyond 1.5 x IQR fences).

Q: Which graph shows distribution shape?

A: Histogram (with optional KDE curve).

Q: Why use heatmap in correlation analysis?

A: Quickly see which variable pairs are strongly correlated.

Q: Which graph is used after PCA?

A: Scatter plot of PC1 vs PC2 to visualize reduced dimensions.

8. Conclusion

Selecting the appropriate graph is essential for effective data analysis. Categorical data is best shown with bar charts; numerical distributions with histograms and box plots; relationships with scatter plots and heatmaps. Our three experiments use these visualizations on the Titanic dataset to explore frequencies, distributions, correlations, and PCA results.